

Peter Morganelli

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EDUCATION

Tufts University

Bachelor of Science in Computer Science

Medford, MA

Expected May 2027

- **Dean's List (6x)**
- **Coursework:** Algorithms, Data Structures, Concurrent Systems, Networks, Natural Language Processing, Machine Structure & Assembly, Intro Cybersecurity

Trinity College Dublin

Semester Abroad, Computer Science

Dublin, Ireland

January 2026 – May 2026

TECHNICAL SKILLS

Languages: C++, C, C#, Python, JavaScript, TypeScript, SQL, x86 Assembly

Frameworks & Libraries: React, React Native, Next.js, Node.js, Django, Spring Boot, Three.js, Unity

Infrastructure & Tools: Docker, AWS (S3, EC2), Vercel, PostgreSQL, MongoDB, Neo4j, Redis, Firebase, Vitest, Git

EXPERIENCE

Software Engineer Intern | *Python, Django, Neo4j, Docker, GitLab*

Bedford, MA

MITRE

May 2026 – Present

- **Parallelized** bounded requirement verification for a **GraphRAG** application with a **four-worker async request pool** handling retries and rate limiting, cutting end-to-end runtime by **77%** on large Excel imports
- Shipped production features on a **Django** requirements platform over a **Neo4j** graph database, maintaining traceability between source requirements, system-model data, and LLM-generated findings for Air Force **CUI** systems
- Presented **AI agent orchestration** and model prompting workflows to **50+ senior engineers**
- Fixed **90+** SonarQube issues, code refactors, security hotspots, and code smells for compliant **DoW** standards

Data Structures Teaching Fellow | *C++, GitLab, Gradescope*

Medford, MA

Tufts University

September 2024 – Present

- Own grading standards for a **200+ student** course, setting the staff-wide call on ambiguous rubric criteria across **300+ submissions**
- Revise project specifications and lecture material with faculty; support students in office hours and on Piazza

Full-Stack Developer Intern | *React, JavaScript, PostgreSQL, MongoDB, AWS*

Medford, MA

Launch

February 2025 – April 2025

- Contributed to an **AI-powered website builder** from scratch at an early-stage startup, working across the **React** frontend and the **RAG** pipeline
- Designed a **drag-and-drop editor** for end users to restructure and customize generated site components

Software Developer | *React, TypeScript, TailwindCSS, Figma*

Medford, MA

Tufts JumboCode

October 2023 – May 2025

- Built a **full-stack** inventory site for a Somerville theater with **10+ developers**, implementing **RESTful API** handlers and **unit tests** across 10+ components
- Shipped an image-upload carousel with per-item metadata and filter tags from a **Figma** prototype

PROJECTS

[Peter's Pizzeria](#) | *React, Vite, Vercel Serverless, Vercel Blob, Upstash Redis, Vitest*

May 2026 – Present

- **Launched** a full-stack ordering platform for my student-run pizzeria with customer checkout, a live kitchen board with **5s polling** configurable store hours, and community image posting across **15 serverless endpoints**
- Hardened the order API against client tampering by recomputing every price **server-side**, authenticating staff with **HMAC-derived tokens** in HttpOnly cookies, and enforcing Redis-backed per-IP **rate limits**
- Wrote a **140-test Vitest suite** that runs on real HTTP handlers, then checked it via mutation testing for enforcement of accurate unit testing

[Programming Languages Interpreter](#) | *Elm, JavaScript, HTML/CSS*

May 2025 – August 2025

- Engineered a zero-setup **web-based IDE** integrating five language interpreters via **Elm ports** and JavaScript bridges for live in-browser execution
- Designed a **modular interpreter API** for adding new languages, built for **200+ students** to complete coursework, under the mentorship of **Professor Emeritus Dr. Norman Ramsey**